Roast my CV: Project Overview and PRD (Draft v0.1)

Domain 1, Aiversecity Future-Ready Workforce, SDC Hack. Solo build. Drafted 20 Oct 2026.

1. Project overview

One-line pitch. Roast my CV is a CV review tool for Malaysian fresh graduates that scores a CV against dated labour-market demand data, then turns the result into ranked future roles, a reskilling pathway and SDG links under two AI-disruption scenarios.

Problem. Domain 1.pdf cites 56% of Malaysian professionals using AI tools at work but only 26% formally trained (Hays Asia Salary Guide 2025), and a Ministry of Science, Technology and Innovation statement that more than 600,000 workers need reskilling within three to five years. Graduates do not know which skills the market is moving toward.

Primary user and moment. A fresh graduate choosing a first career.

Why it is not a chatbot. The roast is the interface. Scores, demand direction and rankings are computed from stored data; the language model only extracts CV content and writes rationales that cite stored sources. Judges penalise "upload a CV and ask a chatbot" (Domain 1.pdf, Innovation criterion, 30%).

Human-centred position. The tool supports career decisions. It never ranks candidates for employers and never makes hiring decisions.

2. SDG alignment

Main SDG: 8. Targets from the brief: 4.4, 4.3, 8.2, 8.5. Each occupation and skill carries at least one SDG tag from a curated mapping table (drafted by the model, with every row for the priority occupations checked by hand). The result page shows how the user's chosen path contributes to those goals.

3. Scope

In scope: CV and form intake, DOSM trend layer, roast, role matching with skill gap, pathway, SDG mapping, two scenarios, EN/BM/ZH interface.

Out of scope: course content, payments, recruitment integrations, employer view (agreed out of scope until the core flow works), real CVs (synthetic only).

Breadth rule (solo constraint). Cover MASCO major groups 1 (Managers, shown as a later-career destination), 2 (Professionals), 3 (Technicians and Associate Professionals), 4 (Clerical Support) and 5 (Services and Sales), with deep skill profiles and pathways for about 30 priority occupations within them. Groups 0, 6, 7, 8 and 9 are out of scope for the prelim. Which groups carry DOSM counts is to be confirmed from the release tables.

4. Key features mapped to the six must-haves

#	Must-have	Delivery	Acceptance criterion
1	Skills profile intake	PDF/DOCX upload plus short form fallback; LLM extracts to a validated JSON schema	Synthetic CV produces skills, education, experience in under 30 seconds
2	AI-trend signal layer	Ingest the DOSM Job Market Insights quarterly release (OpenDOSM API if the series is in the catalogue, otherwise parsed release tables); store snapshots with source URL and date; compute quarter-on-quarter change per occupation group	UI shows source and date; switching or refreshing a snapshot visibly changes rankings
3	Future-role matching	RunPod embedding similarity between CV skills and the occupation skill profile, combined with weighted overlap and adjusted by demand direction	Each match shows score, matched skills, missing skills and the demand evidence

#	Must-have	Delivery	Acceptance criterion
4	Personalised pathway	Ordered steps with estimated time, chosen only from a curated, URL-verified free or low-cost resource table	No step without a source URL; model cannot add resources
5	SDG mapping	Curated occupation-to-SDG table; skill tags inherited	Every role and skill shows at least one SDG; path-level contribution summary
6	Handling uncertainty	Each occupation carries an AI exposure score (0-10 with a written rationale, from a fixed rubric). Moderate and fast disruption scenarios set how strongly exposure reshapes demand, applied to the DOSM demand direction	Both scenarios shown side by side, labelled as estimates; ranking can differ

Roast layer. CV score from a rubric: rule checks (sections present, quantified results, length) plus a model-written critique grounded in the demand data. Tone selector: gentle, standard, blunt. Output language follows the interface language (English, Bahasa Malaysia, Chinese).

5. Architecture and AI usage

- **Frontend:** React (Vite), i18n dictionaries for three languages, Future Skills Radar drawn as a treemap (area = DOSM job offerings, colour layers = demand change, AI exposure, SDG).
- **Backend:** FastAPI with Pydantic schemas; scheduled ingestion job; Postgres.
- **Generation model:** DeepSeek V4.1 Flash via OpenCode Zen (OpenAI-compatible, base URL opencode.ai/zen/v1, model ID deepseek-v4.1-flash). Free-tier quota and terms to be verified. Declared in the Tech and Tools Declaration.
- **Generation uses:** (a) CV extraction, (b) roast text, (c) batch generation of occupation skill profiles, marked "model-estimated", (d) match rationales, (e) batch re-scoring of occupations after each new DOSM snapshot. One model for all five, so score changes

come from the data and not from a model swap. Roast and rationale requests receive a compiled knowledge-context file (occupation, demand direction, exposure score and rationale, skills, SDG, source and date), regenerated after each snapshot.

- **Embeddings (RunPod):** an open embedding model on a RunPod Serverless endpoint maps CV skills to taxonomy skills by similarity. Runpod's worker-infinity-embedding image serves sentence-transformers models through an OpenAI-compatible embeddings route (github.com/runpod-workers/worker-infinity-embedding). Model to be chosen and tested on Bahasa Malaysia and Chinese skill phrases. This endpoint provides the RunPod proof (endpoint or Pod ID, GPU type, runtime).
- **Fallback:** a RunPod-served open chat model through a vLLM endpoint, only if the OpenCode free quota fails during judging and time allows.
- **Hosting:** Replit is compulsory for every team (Online Orientation.pdf).
- **Guardrails:** structured outputs, skills restricted to a fixed taxonomy, no claim without a stored source id, resources only from the curated table.

6. Data sources

- DOSM Big Data Analytics: Job Market Insights, a quarterly release of online job offerings by major occupation group, occupation and economic activity (dosm.gov.my release pages). Latest: Q2 2026, 555,509 offerings, down 11.4 per cent on the quarter. DOSM states earlier editions do not cover all vacancies in Malaysia and must be read with caution. Catalogue: open.dosm.gov.my. Reuse requires the attribution "Source: Department of Statistics Malaysia".
- MASCO 2020 as the occupation taxonomy (DOSM labour force releases).
- Reference pattern: karpathy/jobs scores each occupation with an LLM rubric (0-10 with rationale) and renders a treemap with colour layers (github.com/karpathy/jobs). Adopted patterns, reimplemented and declared rather than copied (no license file was visible in the repo listing): rubric exposure score with rationale, treemap with colour layers, a compiled context file for grounding, the scrape-parse-score-build pipeline order, and the disclaimer that exposure does not mean a job disappears.
- Synthetic CVs only (Domain 1.pdf rules).

7. Risks and fallbacks

Risk	Fallback
Job-offering series not in the OpenDOSM API catalogue	Parse the quarterly release tables into dated snapshot files; pipeline still records source URL and date
Model invents skills or courses	Fixed taxonomy and curated resource table; reject outputs without source ids
Cold start delays the demo	Keep one warm worker during recording; cache last results
Scope too large for one builder	Depth tiers (30 deep occupations); i18n for UI strings first, model output language second
Weak matching quality	Test against 5 to 8 synthetic CVs and tune weights
Treemap with three colour layers is heavy for one builder	Ship the demand layer first, then exposure, then SDG; fall back to a ranked list if the treemap slips

8. Timeline

Deadline: 25 Oct 7:59:59 AM MYT, no late submissions (Online Orientation.pdf). Target submission: night of 24 Oct.

- 20 Oct: repo with Apache 2.0 license, Replit deployment, RunPod endpoint, DOSM ingestion.
- 21 Oct: CV extraction and roast.
- 22 Oct: matching, scenarios, SDG table.
- 23 Oct: pathway, resource table, i18n.
- 24 Oct: synthetic CVs, polish, pitch video (5-7 minutes), declaration, submit, open every link in incognito.

9. Submission checklist

Product Requirement Doc, pitch deck, live prototype link, Tech and Tools Declaration, public GitHub repository, YouTube pitching video. RunPod proof (Pod ID or endpoint, GPU type, runtime) and Replit proof go in the declaration. Genuine commit history; no exposed API keys.

10. Open items to confirm

- Embedding model choice for RunPod, and DeepSeek quality in Bahasa Malaysia and Chinese.
- Dataset ID of the job-offering series in the OpenDOSM catalogue; not found in public search, so check the catalogue directly.
- Whether a hosted API (DeepSeek on OpenCode Zen) satisfies the RunPod rule; confirm with organisers at sdc.xmumhack@gmail.com.
- Scoring weights for the CV rubric and match score.